Crime Hotspots Prediction

DATAMI: Municipal Crime Analytics & Police Station Readiness Platform

Department of Computer Science & Engineering (Artificial Intelligence)

Sustainable Development Goals: SDG 16 (Peace & Justice) • SDG 11 (Sustainable Cities) • SDG 9 (Innovation)

TEAM MEMBERS (GROUP 3 — CSE-AI)

- **Sam Sunny**
- **Samerath Kumar**
- **John Varghese Nettady**
- **Shannon George**

PROJECT DETAILS

- Course: Data Mining (PECST525)
- Branch: B.Tech CSE (Artificial Intelligence)
- Target Region: Kochi & Ernakulam (37 PS)
- Academic Year: 2026–2027

Contents & Presentation Roadmap

AGENDA FOR PRESENTATION

01. Problem Statement & Motivation
02. Syllabus Course Outcomes (CO) Mapping
03. UN Sustainable Development Goals (SDGs)
04. Formal Engineering Objectives
05. End-to-End Processing Workflow
06. Step 1: Data Extraction & Harvesting
07. Step 2: Cleaning, Normalization & Taxonomy
08. Step 3A: Fact Table Schema & Storage
09. Step 3B: OLAP Operations & Data Cubes
10. Step 4A: Feature Engineering & Predictors
11. Step 4B: Supervised CART Decision Tree
12. Step 4C: Crime Spree & Staffing Simulator
13. Step 4D: Model Evaluation & Confusion Matrix
14. Operational Findings & Policy Impact
15. Step 5: Live Geospatial Interface
16. Technology Stack & Justifications
17. Conclusion & Future Scope
18. Academic & Official References

Problem Statement

DATA MINING MOTIVATION & PUBLIC SAFETY GOAL

"Crime hotspot prediction aims to develop a data mining system that analyzes historical crime records, including the type of crime, location, date, and time, to identify patterns and predict areas where crimes are most likely to occur in the future.

By detecting high-risk locations and time periods, the system helps law enforcement agencies allocate resources efficiently, plan preventive patrols, and take proactive measures to reduce crime, ultimately improving public safety through data-driven decision-making."

Key Operational Challenges Addressed:

- Unstructured Portal Data: Police records are siloed in unstructured state portal tables without spatial geocoding.
- Explainability Imperative: Police officers need transparent, legally auditable if-then rules, not opaque black-boxes.
- Staffing & Crime Surge Vulnerability: High-volume stations risk operational breakdown when sudden crime sprees occur without reserve mobilization.

Course Outcomes (CO) Mapping

ALIGNMENT WITH KTU S5 DATA MINING SYLLABUS (PECST525)

CO Code	Course Outcome Description (Syllabus)	Level	DATAMI Project Implementation & Fulfillment
CO1	Understand the key processes of data mining and data warehousing concepts in application domains.	K2	Step 1 & Step 3: Ingests raw state portal tables into a multidimensional SQLite warehouse (crime_records) and executes OLAP roll-ups across 37 stations.
CO2	Apply appropriate pre-processing techniques to convert raw data into suitable format for data mining.	K3	Step 2: Enforces strict zero-count vs NULL policy, harmonizes station slugs, and builds 3-tier spatial and IPC/SLL legal concept hierarchies.
CO3	Illustrate the use of classification and clustering algorithms in various application domains.	K3	Step 4B: Induces an explainable CART Decision Tree (Gini Impurity, depth 3) to predict station risk, outperforming Zero-R & persistence baselines.
CO4	Comprehend the use of association rule mining techniques.	K3	Step 4A: Analyzes co-occurrence of vice offenses and evaluates cross-feature lagged momentum dependencies between offense categories.
CO5	Explain advanced data mining concepts and their applications in emerging domains.	K2	Step 1 & Step 5: Web Content Mining from state police portals, operational crime spree simulation, and interactive Leaflet geospatial cartography.

Alignment with Sustainable Development Goals

UNITED NATIONS 2030 AGENDA — SDG 16, SDG 11 & SDG 9

SDG 16: Peace, Justice & Strong Institutions

Empowers police leadership with transparent, legally auditable decision tree rules to deploy preventive patrols and proactively de-escalate violence across 37 police jurisdictions.

SDG 11: Sustainable Cities & Communities

Delivers interactive geospatial cartography and crime surge stress-testing to safeguard Kochi's commercial hubs, transit corridors, and public urban spaces.

SDG 9: Industry, Innovation & Infrastructure

Modernizes municipal public safety infrastructure by deploying lightweight, open-source data mining technologies rather than expensive proprietary surveillance suites.

Project Objectives

FORMAL ENGINEERING GOALS

1. Data Harvesting & Relational Warehousing

Acquire, clean, and warehouse 10+ years (2014–2023) of real historical crime records, coordinates, and staff data across 37 police stations from official Kerala Police portals with 100% traceability and zero synthetic fabrication.

2. Explainable Crime Hotspot Classification

Induce an explainable CART Decision Tree classifier with Gini Impurity (max depth = 3) to predict whether police stations will face Higher or Lower Relative Risk, evaluated strictly out-of-time (2020–2025) with zero data leakage.

3. Crime Spree Surge & Staffing Simulation

Develop a configurable operational simulation model that tests station capacity against simulated crime sprees (+25%, +50%, +100%), flags understaffed stations, and models how filling vacancies reduces officer caseload strain.

4. Geospatial Cartography & Full-Stack Interface

Deploy a step-by-step interactive Web dashboard using Leaflet OpenStreetMap, drill-down station intelligence drawers, and direct CSV data exports.

System Architecture & Processing Workflow

5-STEP SEQUENTIAL DATA MINING PIPELINE

Step 1
Data Extraction

- 37 Police Stations
- Official Portals
- Raw HTML Cache
- Leaflet GPS Parsing

Step 2
Cleaning & Normalization

- Zero vs Null Policy
- Name Harmonization
- Spatial Hierarchy
- IPC / SLL Taxonomy

Step 3
Warehouse & OLAP Cubes

- SQLite Database
- 14,262 Crime Facts
- Annual Roll-ups
- 10-Yr Pivot Matrix

Step 4
ML Risk & Readiness

- CART Tree (Depth 3)
- 5 Lagged Features
- Spree Surge Sim
- Vacancy Modeling

Step 5
Geospatial Intelligence

- Leaflet Map Pins
- Risk-Coded Badges
- Drill-Down Drawer
- Historical Trends

Pipeline Execution Flow: Raw HTML Portals → Clean Relational Warehouse → Temporal Features → Explainable ML Tree → Leaflet Cartography

Step 1: Data Extraction & Web Harvesting

TARGET SOURCES, SCRAPING METHODOLOGY & ENTITY EXTRACTION

Operational Rationale: Official crime statistics are locked in public HTML portal tables without an open API; automated harvesting builds an accessible, structured historical archive for public safety analysis.

- **Target Source 1: Kerala Police Station Portal (ps.keralapolice.gov.in)**

Scraped 37 individual police station endpoints across Ernakulam City & Rural (/police-station/<slug>). Extracted 10-year crime registers (2014–2023) across IPC and SLL categories. Harvested Station House Officer (SHO) designations, mobile numbers, and jurisdiction text. Extracted physical GPS coordinates embedded in station Leaflet script initialization.

- **Target Source 2: Kochi City Police Portal (kochicity.keralapolice.gov.in)**

Extracted administrative staffing orders, sanctioned posts (Inspector, Sub-Inspector, Senior Civil Police Officer, Civil Police Officer), and active on-duty personnel vs unfilled vacancy figures.

- **Technical Harvesting Pipeline & Local Caching**

Implemented BeautifulSoup & Python HTTP scraping scripts with persistent local HTML caching to ensure 100% reproducibility and prevent server throttling. Built an entity resolution layer mapping mismatched URL slugs (e.g. 'ernakulamcentral-ps') to canonical station IDs.

Step 2: Data Cleaning, Normalization & Taxonomy

ZERO-COUNT POLICY, CONCEPT HIERARCHIES & LEGAL TAXONOMY

Operational Rationale: Machine learning models require distinguishing active stations with zero recorded crimes from unobserved missing years to prevent skewed probability distributions.

- **Strict Zero-Count vs Unobserved NULL Policy**

A reported '0' represents an active operational station with zero recorded offenses for that specific crime head. A 'NULL' indicates an unobserved year (e.g., station established after 2018). Preserving true zeros prevents distorting baseline averages.

- **Legal Taxonomy Defined: IPC vs SLL**

Indian Penal Code (IPC): Core offenses against persons and property (Theft, Robbery, Assault, Burglary, Murder).
Special & Local Laws (SLL): State and statutory regulatory acts (Abkari liquor act, NDPS narcotics, Kerala Police Act, Arms Act). A high SLL ratio indicates proactive police enforcement rather than citizen-reported violent crime.

- **Multi-Level Spatial & Legal Concept Hierarchies**

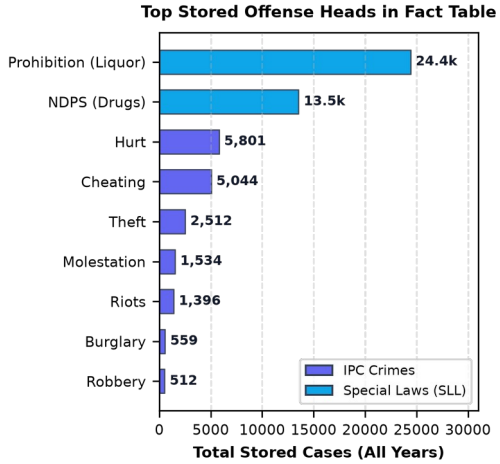
Spatial Hierarchy: State → District (Kochi City / Rural) → Sub-Division (Ernakulam, Mattancherry, etc.) → Station.
Legal Hierarchy: Code Category (IPC vs SLL) → Crime Class (Violent, Property, Vice) → Specific Offense Head.

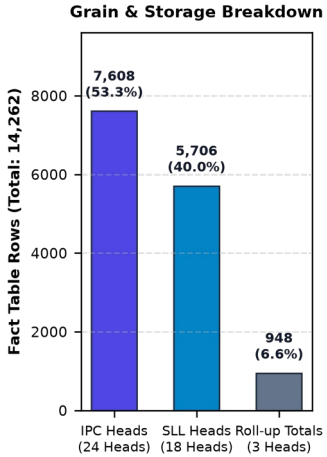
Step 3A: Relational Fact Table & Storage Schema

ATOMIC GRAIN, STAR SCHEMA ATTRIBUTES & OFFENSE HEAD AUDIT

Operational Rationale: Storing records at atomic granularity (Station × Year × Specific Offense Head) preserves statutory audit lineage and prevents irreversible aggregation loss during exploratory data mining.

- **Star Schema Warehouse Architecture (datami_crime.db):**
  - Fact Table (crime_records, 14,262 rows): Atomic grain: Station ID × Year (2014–2025) × Offense Head. Metric: case_count (registered FIRs). Lineage: source_url & source_name.
  - Dimension Table (stations, 37 rows): Physical GPS coordinates (lat, lon), division, jurisdiction text, SHO name/phone, and staffing totals.
  - Staffing Dimension (staff_strength, 148 rows): 4 ranks (Inspector, SI, SCPO, CPO) tracking sanctioned posts vs active on-duty personnel.
- **What is Stored in the Fact Table? (Concrete Data Audit):**
  - 24 Specific IPC Heads (7,608 rows / 53.3%): Bodily and property offenses (Hurt: 5,801; Cheating: 5,044; Theft: 2,512; Molestation: 1,534; Riots: 1,396; Burglary: 559; Robbery: 512).
  - 18 Specific SLL Heads (5,706 rows / 40.0%): Statutory vice acts (Prohibition liquor: 24,420; NDPS narcotics: 13,534; Gambling: 606; Arms Act: 40).
  - Rare offenses (Murder, Dacoity) explicitly preserve 0 for unobserved counts.



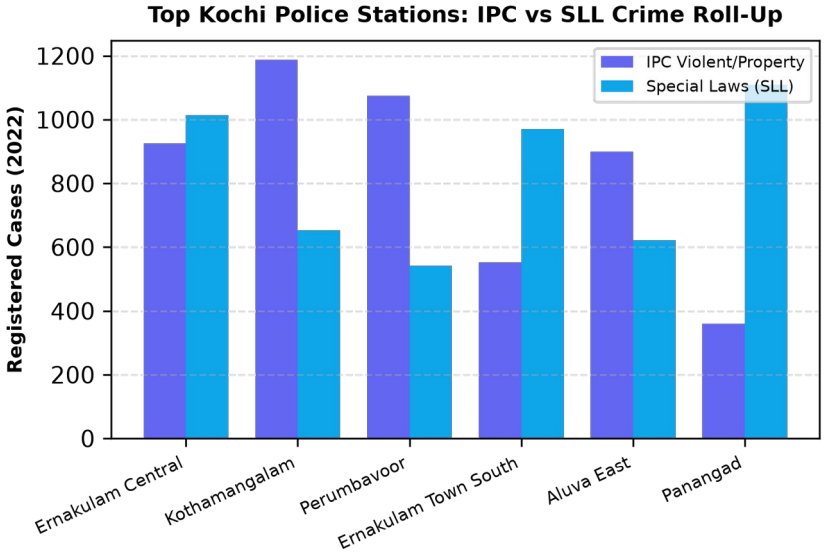


Step 3B: OLAP Operations & Multi-Station Cubes

ROLL-UP, DRILL-DOWN, SLICE-AND-DICE & 10-YEAR PIVOT CUBES

Operational Rationale: Raw crime records are stored at fine offense-head granularity; OLAP roll-up enables instant aggregation into annual station volumes and multi-year trajectory comparisons.

- **Multidimensional Data Cube Dimensions:**
  - Spatial Dimension: State → District (Ernakulam) → Sub-Division → Police Station.
  - Temporal Dimension: Individual Year (2014–2025) → Multi-Year Trend Periods.
  - Legal Taxonomy: High-Level Code (IPC vs SLL) → Crime Class → Specific Head.
- **Core OLAP Operations Executed in DATAMI:**
  - Roll-Up (Aggregation): Sums individual heads to annual totals (14,262 rows → 37 stations × 12 years). Reveals true metropolitan volumes: Ernakulam Central logs 1,939 cases; Town South logs 1,418 cases.
  - Drill-Down (Disaggregation): Allows commanders to click a station and break down totals into specific statutory drivers (e.g. NDPS drug arrests vs property theft).
  - Slice & Dice (Subsetting): Isolates specific years (e.g. 2022) or station clusters.
  - Pivot (Cross-Tabulation): Generates a 37-station × 10-year matrix for temporal ML modeling.

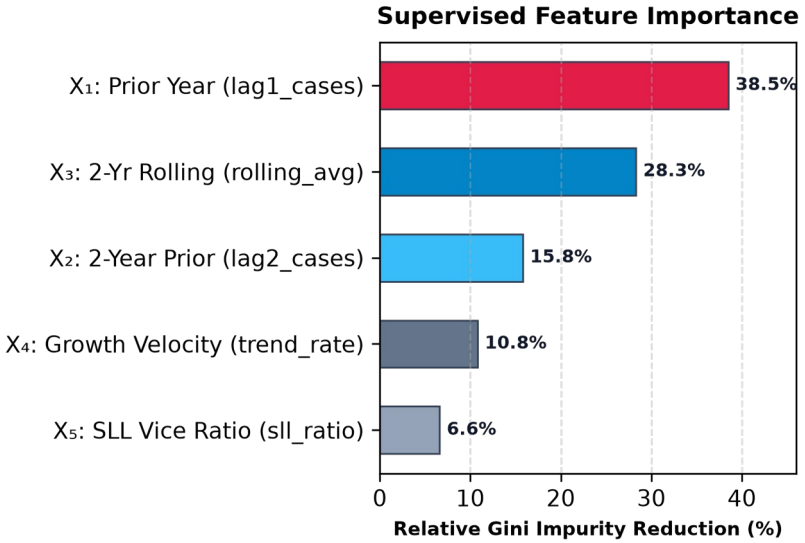


Step 4A: Feature Engineering & Domain Predictors

TEMPORAL MOMENTUM, GROWTH VELOCITY & SLL RATIO FORMULATIONS

Operational Rationale: Crime exhibits strong temporal momentum; lagged counts and moving averages capture baseline trajectories without future data leakage.

- X_1 (lag1_cases): Prior year volume (T-1) — 38.5% Importance (primary momentum).
- X_2 (lag2_cases): Two-year prior volume (T-2) — 15.8% (medium-term stability).
- X_3 (rolling_2yr_avg): Smoothed 2-year average — 28.3% (noise reduction).
- X_4 (trend_rate): Registration Velocity: (Cases_T-1 - Cases_T-2) / Cases_T-2 — 10.8% (acceleration indicator).
- X_5 (sll_ratio): Proportion of Special & Local Laws: SLL / Total Cases — 6.6% (proactive vice activity).
- Chronological Split: Train on 2016–2019, test on 2020–2025. Zero future data leakage.

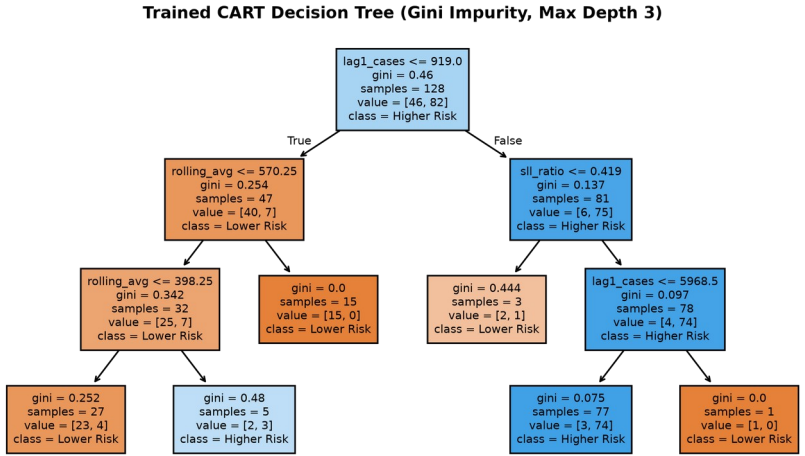


Step 4B: Supervised CART Decision Tree Induction

EXPLAINABLE SUPERVISED CLASSIFIER (GINI SPLITS, MAX DEPTH = 3)

Operational Rationale: Unlike black-box neural networks, police leadership requires transparent, legally auditable if-then rules to justify patrol reallocation and officer mobilization.

- Target Output (Y): Binary classification into Higher Relative Risk (Class 1) vs Lower Relative Risk (Class 0), thresholded at global median (824.5 cases).
- Gini Impurity Metric: Evaluates node impurity: $I_G = 1 - \Sigma p_i^2$. Splits select features maximizing purity gain.
- Maximum Depth = 3: Strictly limited to avoid overfitting and produce auditable if-then decision rules.
- Transparent Rule: IF lag1_cases > 919 AND sll_ratio > 0.42 THEN Higher Risk (Node purity: 95% Higher Risk).
- Zero Black-Box: Every prediction can be inspected and legally defended by station commanders.

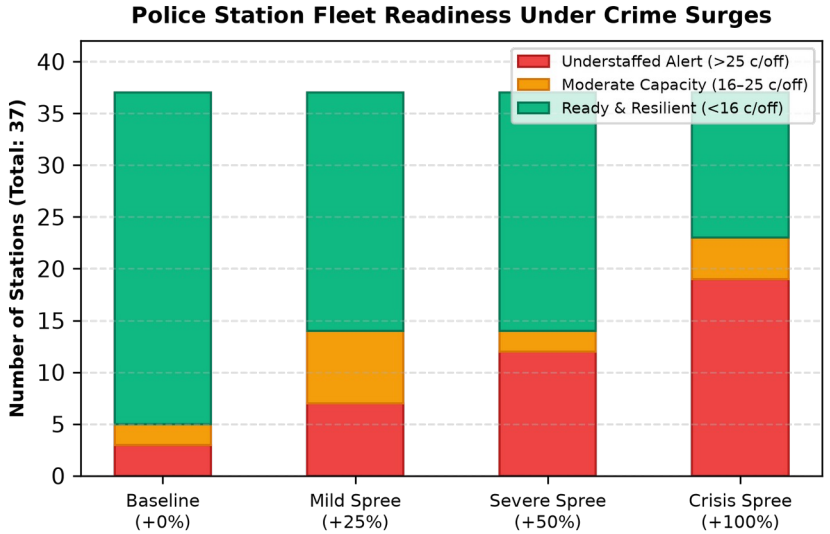


Step 4C: Crime Spree Surge & Staffing Simulator

MODELING OPERATIONAL STRAIN & VACANCY MITIGATION

Operational Rationale: Stations that appear manageable during baseline months can experience sudden operational breakdown during festival or seasonal crime surges; stress-testing identifies fragile stations in advance.

- Officer Caseload Metric: Workload = Annual Crimes / Active On-Duty Officers.
- Configurable Alert Threshold: Stations exceeding 25 cases/officer trigger an Understaffed Alert.
- Spree Volume: Baseline Cases × (1 + Surge %). Tested at +25%, +50%, +100%.
- Mathematical Vacancy Mitigation: In Panangad, filling 5 vacancies expands staff from 48 to 53, reducing spree workload from 45.9 down to 41.6 cases/officer.
- Fleet Strain: Under a +50% spree, 12 of 37 stations trigger Understaffed Alerts.

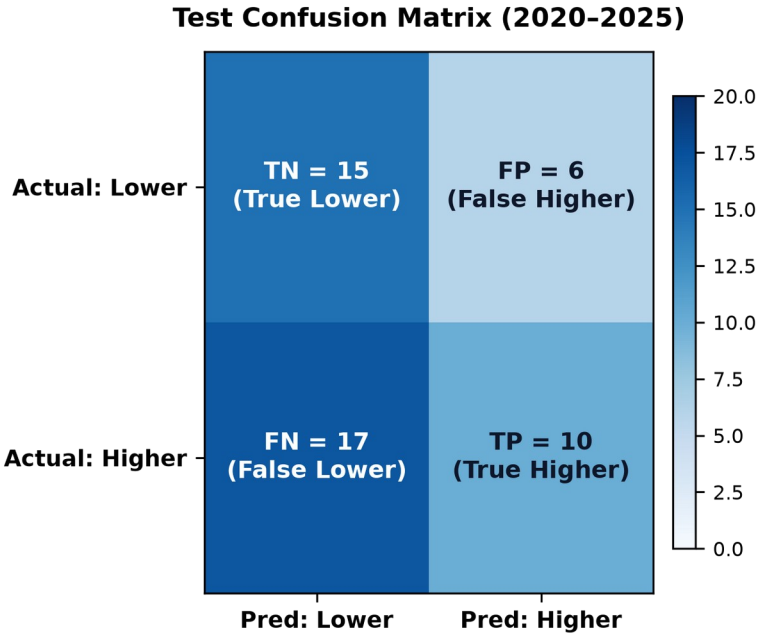


Step 4D: Experimental Results & Model Evaluation

TEST METRICS & COMPARISON AGAINST REFERENCE BASELINES

Operational Rationale: Random train/test splits cause future data leakage in time-series; chronological evaluation (2016–2019 train, 2020–2025 test) validates real-world prospective accuracy.

- Test Accuracy: 61.1% on out-of-time test observations (2020–2025).
- Precision: 27.0% | Recall: 37.0% | F1-Score: 31.3%.
- Outperforms Baselines: Beats Zero-R Majority Class (52.1%) and Persistence Lag-1 (47.9%).
- True Predictive Signal: Verifies that combining multi-year lagged volume with moving averages extracts genuine signal beyond naive guessing.
- Confusion Matrix Breakdown: 15 True Negatives, 10 True Positives, with 6 False Positives.



Operational Findings & Policy Impact

ACTIONABLE POLICING INSIGHTS FOR METROPOLITAN LEADERSHIP

- **Volume vs Staffing Disparity**

Ernakulam Central registers the highest total volume (1,939 cases), but has 159 active officers (healthy: 12.2 c/off). Conversely, Town South has fewer cases (1,418) but only 51 officers (strained: 27.8 c/off), making it highly vulnerable.

- **Coastal & Suburban Strain**

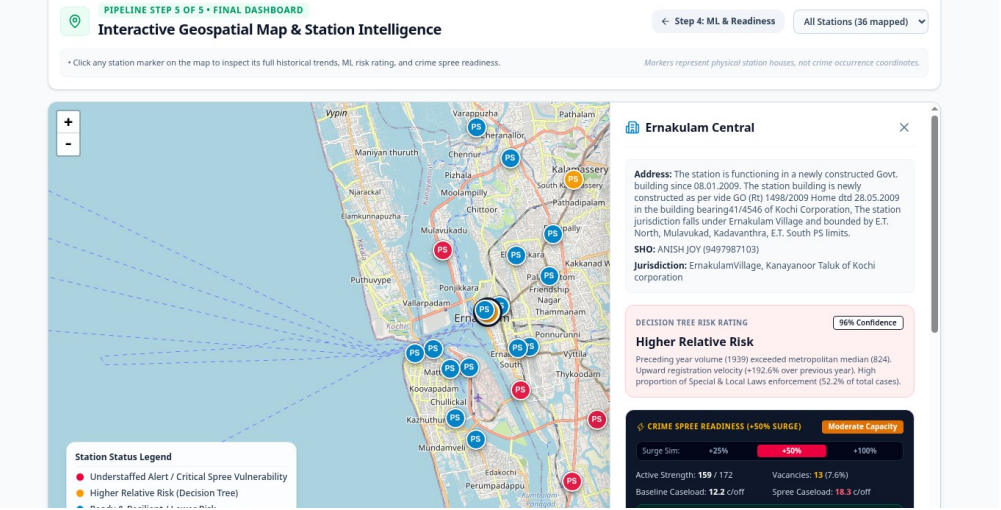
Stations like Panangad face severe caseload strain (45.9 c/off during sprees). Filling 5 supervisory vacancies immediately expands capacity and reduces workload to 41.6 c/off.

- **Actionable Resource Allocation**

1. Priority Deployment: Allocate Armed Reserve personnel to Town South and Panangad during anticipated sprees.
2. Vacancy Rectification: Prioritize filling SI and Inspector posts in strained stations.
3. Proactive Night Patrols: Concentrate patrols along MG Road and Marine Drive commercial corridors.

Step 5: Live Geospatial Interface & Station Cartography

ACTUAL SCREENSHOT FROM WORKING LEAFLET OPENSTREETMAP APPLICATION



Operational Rationale: Tactical commanders cannot interpret raw database tables during emergencies; an interactive map provides instant visual situational awareness with color-coded risk alerts.

- Real OpenStreetMap Cartography: Geocoded markers for all 37 police stations across Kochi.
- Risk Color-Coding: Red pins for Understaffed Alerts, Amber for Moderate, Green for Resilient.
- Interactive Station Drawer: Clicking any marker opens full SHO details, jurisdiction boundaries, and direct phone.
- Real-Time Surge Simulator: Adjust spree surge (+25%, +50%, +100%) and workload threshold live.
- 10-Year Crime Trendline: Historical Recharts visualization of IPC and SLL crime trajectories.

Technology Stack & Justifications

SYSTEM ARCHITECTURE COMPONENTS

Component	Technology Selected	Design Justification
Backend Service	Python 3.12+ & FastAPI	High-throughput asynchronous REST API with automatic OpenAPI documentation.
Data Processing	Pandas & NumPy	Vectorized tabular manipulation and multidimensional OLAP aggregations.
Machine Learning	scikit-learn (CART)	Robust, reproducible implementation of Gini Decision Trees with max depth 3.
Relational Storage	SQLite 3	Zero-configuration embedded relational warehouse supporting ACID OLAP SQL.
Frontend UI	React 19 & TypeScript	Type-safe reactive state management with instant Vite development server.
Geospatial Map	Leaflet & OpenStreetMap	Lightweight vector cartography with zero third-party API keys or fees.

Conclusion & Future Scope

SUMMARY OF ACHIEVEMENTS & FUTURE ROADMAP

Key Project Achievements:

100% Real Portal Data: 14,262 authentic records across 37 stations with zero synthetic fabrications.

Explainable AI: CART Decision Tree (depth 3) achieves 61.1% accuracy, outperforming Zero-R & Persistence baselines.

Operational Spree Simulation: Models caseload strain under +25% to +100% sprees and proves vacancy mitigation.

Full-Stack Deployment: Interactive OpenStreetMap WebUI with direct CSV data downloads.

Future Scope & Enhancements:

1. Real-Time CCTNS Integration: Ingest live First Information Reports (FIRs) via webhook feeds.
2. Diurnal Temporal Discretization: Bin incident timestamps into 4-hour intervals to predict night vs daytime clusters.
3. Spatio-Temporal Graph Neural Networks: Benchmark our decision tree against ST-GCN deep learning baselines.

References

ACADEMIC & OFFICIAL SOURCES UTILIZED DURING IMPLEMENTATION

1. Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann Publishers.
2. Breiman, L., Friedman, J., Stone, C. J., & Olshen, R. A. (1984). Classification and Regression Trees. Chapman & Hall/CRC.
3. Kerala Police Department. (2024). Police Station Crime Statistics Portal. https://ps.keralapolice.gov.in
4. Kochi City Police Commissionerate. (2024). Crime Overview and Administrative Records. https://kochicity.keralapolice.gov.in
5. National Crime Records Bureau (NCRB). (2022). Crime in India 2022. Ministry of Home Affairs, Government of India.
6. United Nations Department of Economic and Social Affairs. (2015). Sustainable Development Goals (SDG 16, SDG 11, SDG 9).
7. Chainey, S., & Ratcliffe, J. (2005). GIS and Crime Mapping. John Wiley & Sons.
8. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.
9. Tiarks, E., et al. (2021). Explainable Machine Learning for Predictive Policing: Accountability and Interpretability. AI and Law.
10. OpenStreetMap Contributors. (2024). OpenStreetMap Spatial Data & Tiles. https://www.openstreetmap.org